SKYWAVE™
AN ORBCOMM COMPANY



ST 8100
Hardware Guide

T416, Version 04
Mar 2026

© ORBCOMM®

CONTACT AND LEGAL INFORMATION

Visit SKYWAVE Online

Website	www.skywave.com
Address	750 Palladium Drive, Suite 368, Ottawa, Ontario CANADA K2V 1C7
Head Office	22970 Indian Creek Drive, Suite 300, Sterling, VA 20166 USA

Contact Partner Support

Portal	https://partner-support.orbcomm.com
Email	partner-support@orbcomm.com
Phone	+1.613.836.2222

Export Control Statement

The contents of this document, in whole or in part, shall not be exported from the United States, which export shall include, but not be limited to, transmittal to any non-U.S. citizen wherever said person is located, except in accordance with all United States laws and regulations relating to exports and to all administrative acts of the U.S. Government pursuant to such laws and regulations. Diversion, re-export or transshipment of the contents of this document, in whole or in part, contrary to U.S. law is also strictly prohibited.

Trademark Notice

The SKYWAVE name and logo are trademarks of ORBCOMM Inc. and may not be used without permission.

ORBCOMM is not responsible for the operation and regulatory compliance of the Inmarsat® satellite system.

Other trademarks, trade names, and logos are those of their respective owners.

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

TABLE OF CONTENTS

Contact and Legal Information **2**

Contact Partner Support 2

Trademark Notice 2

Preface **5**

Purpose **5**

Notation **5**

Reference **5**

Battery Safety Warnings **5**

WEEE Statement **6**

1 Product Overview **7**

2 Specifications **9**

2.1 Temperature **9**

2.1.1 Internal Battery Temperature **9**

2.2 Electrical **9**

2.2.1 Power **9**

2.3 RF Specifications **9**

2.3.1 OGx **9**

2.3.2 Satellite Antenna **10**

2.3.3 Cellular Antenna **10**

2.4 BLE Antenna **10**

2.5 ST 8100 EIRP **10**

2.6 GNSS Module **11**

2.7 Battery **11**

2.7.1 Battery Voltage Measurement **11**

2.7.2 Battery Electrical **11**

2.7.3 Battery Charging **11**

2.7.4 Battery Controller **12**

2.8 Memory **12**

2.8.1 Non-Volatile Flash Storage **12**

2.8.2 External Memory **12**

2.9 Sensors **12**

2.9.1 Temperature Sensor **12**

2.9.2 Accelerometer **13**

2.10 BLE **13**

2.11 Cellular Radio **13**

2.12 LED **13**

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

2.13 Mechanical 14
2.14 Advanced Variant Features 14
2.14.1 Connectors (Advanced Variant) 14
2.14.2 Configurable I/Os (Advanced Variant) 15
2.14.2.1 Digital Input 16
2.14.2.2 Digital Output 17
2.14.2.3 Analog Input 18
2.14.3 Serial Interfaces (Advanced Variant) 18
2.14.3.1 RS-232 18
2.14.3.2 RS-485 19
2.14.4 1-Wire (Advanced Variant) 19
2.14.5 Open Cover Detection (Advanced Variant) 19
2.14.6 24 V Output (Advanced Variant) 20
2.14.7 High Voltage Analog Input (Advanced Variant) 20
2.14.8 Analog Current Input (Advanced Variant) 20
3 Compliance 21

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

PREFACE

Purpose

This guide is as an overview of the hardware characteristics and specifications for the ST 8100 (Standard and Advanced variant). The information in this guide refers to both variants unless noted otherwise.

Notation

Hardware components and hardware labels in this document might not be exactly as shown and are subject to change without notice.

CAUTION: This safety symbol warns of possible hazards to personnel, equipment, or both. It includes hazards that will or can cause personal injury, property damage, or death if the hazard is not avoided.

Note: A note indicates information with no potential hazard. A note indicates points of interest or provides supplementary information about a feature or task.

Bulleted lists highlight information where order or sequence is not crucial.

Reference

The content of the following documents might be useful in conjunction with this guide.

Document names and numbers are subject to change, or be discontinued, without notice.

[T404] LSF Developer Guide

[T405] LSF API Reference Guide

[T417] ST 8100 Installation Guide

[N206] MTWS Cellular Protocol

[N214] OGx Gateway Web Service (OGWS) User Guide

Battery Safety Warnings

CAUTION: DO NOT short circuit or expose the battery to temperatures above the maximum rated temperature.

CAUTION: Always follow local disposal guidelines to properly dispose of the Lithium-ion battery and the device.

CAUTION: Store in a cool, well-ventilated area. Elevated temperatures will result in shortened battery life.

CAUTION: DO NOT throw the internal battery or the device into fire.

CAUTION: DO NOT replace the battery. Changing the battery without ORBCOMM's permission could violate regulatory conformity.

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

WEEE Statement

Disposal of this device should be handled according to all national laws and regulations.

The mark shown to the right is in compliance with Waste Electrical and Electronic Equipment Directive 2012/19/EU (WEEE). The mark indicates the requirement NOT to dispose the equipment as unsorted municipal waste but use the return and collection systems according to local law.	

1 PRODUCT OVERVIEW

The ST 8100 (model number ST8100) dual-mode satellite-cellular terminal supports cost-effective IoT solutions that track and monitor mobile and fixed assets in industries such as maritime, transportation, oil and gas, agriculture, water management and more. It is ideal for applications without external power, such as tracking small vessels or monitoring fixed assets in remote locations, thanks to its solar-powered battery.

The ST 8100 has two variants:

- Standard - no external connector and basic functionality (part number ST8100-0100-H)



- Advanced - a 16-pin external connector and open cover detection (part number ST8100-1100-H)



The ST 8100 Standard Variant includes the following features and functionality:

- LED to indicate overall health
- Internal rechargeable battery
- GNSS with support for GPS, GLONASS, BeiDou, Galileo, and BeiDou 1I
- Bluetooth Low Energy (BLE)
- Motion detection accelerometer
- and more

The ST 8100 Advanced Variant includes all the features and functionality of the standard variant and also includes the following:

- 16-pin connectors for direct integration with sensors
- Configurable I/O ports
- Serial interfaces (RS-232 and RS-485)
- Open cover detection

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

- 24 V Output
- High Voltage Analog Input
- Analog Current Input

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

2 SPECIFICATIONS

2.1 Temperature

Parameter	Value
ST 8100 Operating Temperature Range	-35° to +70°C (-31°F to +158°F)
ST 8100 Storage Temperature Range	-40° to +85°C (-40°F to +185°F)

2.1.1 Internal Battery Temperature

The internal battery's temperature specifications are:

Parameter	Value
Internal Battery Operating Temperature Range	-35°C to +70°C (-31°F to +158°F)
Internal Battery Charging Temperature Range (Standard)	-20°C to +70°C (-4°F to +158°F)
Internal Battery Charging Temperature Range (Advanced)	-35°C to +70°C (-31°F to +158°F)

2.2 Electrical

2.2.1 Power

The ST 8100 is powered through the internal battery that is recharged using an integrated solar panel.

2.3 RF Specifications

The device includes two-way satellite communication to interface with the OGx networks.

2.3.1 OGx

Parameter	OGx
Receive	
Frequency Band	1525 to 1559 MHz
Modulation	OQPSK
Symbol Rate	2000, 8000, 16000 symbols / second
Polarization	RHCP
Transmit	
Frequency Band	1626.5 to 1660.5 MHz
Modulation	OQPSK
Symbol Rate	800, 1600, 3200, 6400, 12800 symbols / second
Polarization	RHCP

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

2.3.2 Satellite Antenna

Parameter	Value
Antenna Type	Patch
Frequency Band	1525 to 1660.5 MHz
Impedance	50 Ω
VSWR (with matching network)	<2
Polarization	RHCP
Antenna Gain	4.2 dBi typical, 5 dBi maximum

2.3.3 Cellular Antenna

Parameter	Value
Frequency Range	698 to 960 / 1710 to 2690 MHz
Impedance	50 Ω
VSWR	≤7.3 for 698 to 960 MHz and ≤2.3 for 1710 to 2690 MHz
Gain (total realized)	3.0 dBi maximum for 698 to 960 MHz 3.0 dBi maximum for 1710 to 2000 MHz 3.4 dBi maximum for 2000 to 2170 MHz 4.0 dBi maximum for 2300 to 2690 MHz
Polarization	Linear

2.4 BLE Antenna

Parameter	Value
Antenna Type	Inverted F
Frequency Range	2.4 to 2.485 GHz
Impedance	50 Ω
VSWR	≤1.9:1
Gain (realized gain)	Phi: 2.7 dBi maximum Theta: 2.6 dBi maximum
Polarization	Linear

2.5 ST 8100 EIRP

The EIRP of the ST 8100 with an elevation of 10° is ≥-3.5 dBW, with a maximum EIRP of +4.5 dBW.

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

2.6 GNSS Module

The major constellations include: GPS, GLONASS, Galileo, BeiDou and BeiDou B1I.

Parameter	GPS	GLONASS	BeiDou B1I	Galileo	BeiDou
Time to First Fix					
Cold Start	29 s	27 s	30 s	41 s	56 s
Hot Start	1 s	1 s	1 s	1 s	1 s
Aided Start	1 s	1 s	1 s	5 s	N/A
Sensitivity					
Tracking and Navigation	-167 dBm	-166 dBm	-160 dBm	-161 dBm	-163 dBm
Cold Start	-148 dBm	-147 dBm	-146 dBm	-141 dBm	-136 dBm
Hot Start	-159 dBm	-159 dBm	-159 dBm	-155 dBm	-157 dBm
Accuracy					
Horizontal Position	1.5 m CEP	4.0 m CEP	2.0 m CEP	3.0 m CEP	2.0 m CEP
Velocity	0.05 m/s				
Heading	0.3 degrees				

2.7 Battery

The battery is a Li-ion rechargeable battery.

2.7.1 Battery Voltage Measurement

The device can measure the input voltage over the range 2.5 V to 4.2 V.

2.7.2 Battery Electrical

The device includes a long life, rechargable battery capable of recharging from the integrated solar panel.

The electrical specifications for the battery are:

Parameter	Min.	Typical	Max.	Units
Capacity	1000	-	-	mAh
Voltage (nominal)	-	3.6	-	V
Current Output	2.0	-	-	A
Charge Current	1	-	200	mA

2.7.3 Battery Charging

The battery charger maximizes power transfer from the solar panel to the battery regardless of available sunlight intensity down to 1 mA of charge current.

The battery charging electrical specifications are:

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

Parameter	Min.	Typical	Max.	Units
Solar Open Circuit Voltage	4.7	-	-	V
Solar Power	770	-	-	mW
Charge current	1	-	125	mA

2.7.4 Battery Controller

Charge and discharge are controlled so that battery discharge is not allowed over the battery's safe discharge operating conditions.

The battery controller electrical specifications are:

Parameter	Typical	Max.	Units
Charge Current	-	200	mA
Float Voltage	4.05	-	V
Charge Time	-	8	Hours
Over Discharge Voltage	2.5	-	V
Over Charge Voltage	4.2	-	V
Over Discharge Current	2.5	-	A
Short Circuit Current	6	-	A

2.8 Memory

2.8.1 Non-Volatile Flash Storage

The device includes 128 Mbit of non-volatile on-board flash storage. The flash is capable of 100,000 write-erase cycles over its operating life. The external flash memory also has small block sizes of 4 KB or less.

2.8.2 External Memory

The device has 64 Mbit of PSRAM external memory.

2.9 Sensors

2.9.1 Temperature Sensor

Parameter	Value
Range	-40 to +85°C (-40 to +185°F)
Accuracy (typical)	±2°C (±3.6°F)

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

2.9.2 Accelerometer

The ST 8100 has an accelerometer to detect motion. To minimize false motion detections and excess power usage, the internal motion thresholds are configurable. Refer to T405 for additional details.

Parameter	Condition	Min.	Typ.	Max.	Units
Acceleration Range	software selectable	-	±2	-	g
		-	±4	-	g
		-	±8	-	g
		-	±16	-	g
Sensitivity	2 g	-	16384	-	LSB/g
	4 g	-	8192	-	LSB/g
	8 g	-	4096	-	LSB/g
	16 g	-	2048	-	LSB/g
Bandwidth Filtering	Selectable via digital interface	12.5	-	1600	Hz

2.10 BLE

The characteristics for the BLE are:

Parameter	Min.	Typical	Max.	Units
Frequency	2402	-	2480	MHz
Bluetooth version compliance	-	5	-	-
Receive sensitivity	-95	-	-	dBm
TX power	-	-	+7.2	dBm
Available RAM	256	-	-	kBytes
Available flash	1	-	-	MBytes
Peripherals available	-	-	4	-

2.11 Cellular Radio

The device is capable of global cellular communications over LTE and 2G networks.

The cellular radio specifications are:

Parameter	Minimum Specification
LTE Category	Category M1 and NB2
LTE Bands	1, 2, 3, 4, 5, 8, 12, 13, 18, 19, 20, 25, 26, 28, 66, 85, and 2G 4-band
GSM Bands	GSM850/EGSM900/DCS1800/PCS1900

2.12 LED

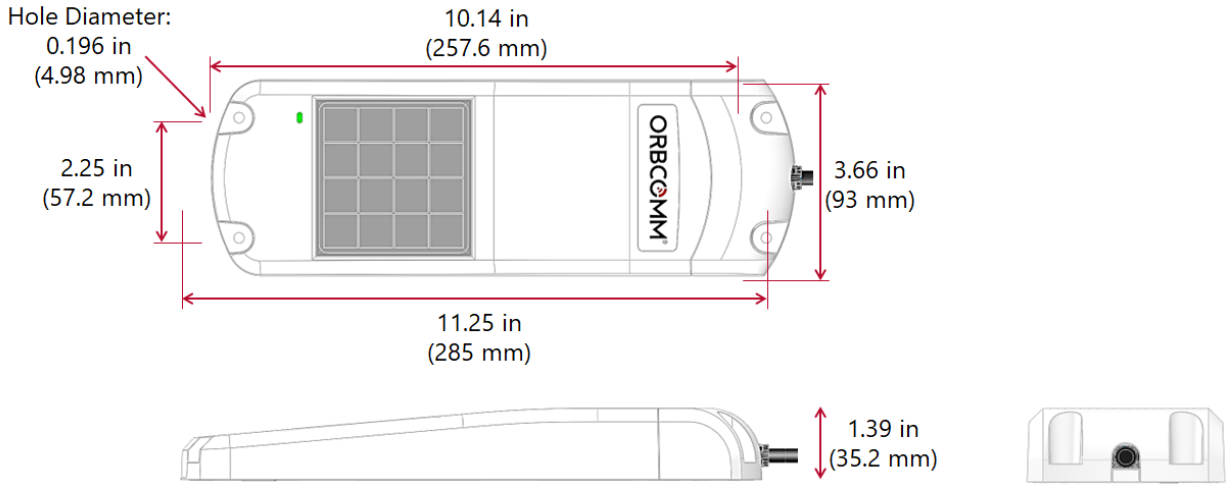
The ST 8100 has a single LED to indicate general health of the device.

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

2.13 Mechanical

Parameter	Value
ST 8100 Dimensions (includes mounting features, excludes connector body)	285 x 93 x 35 mm (11.3 x 3.7 x 1.4 in)
ST 8100 Weight Standard (no connector) Advanced (with connector)	 485 g (17 oz) 500 g (17.6 oz)

Figure 1: ST 8100 Dimensions (Advanced Variant shown)



2.14 Advanced Variant Features

2.14.1 Connectors (Advanced Variant)

The advanced variant has a 16-pin connector. The connector allows various peripherals such as RS-232/RS-485, configurable I/O, high voltage analog inputs, and analog current inputs. The cable harness to interface with the connector is part number: ST101697-00X.

Table 1 maps to the layout shown in Figure 2.

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

Figure 2: View of Connector

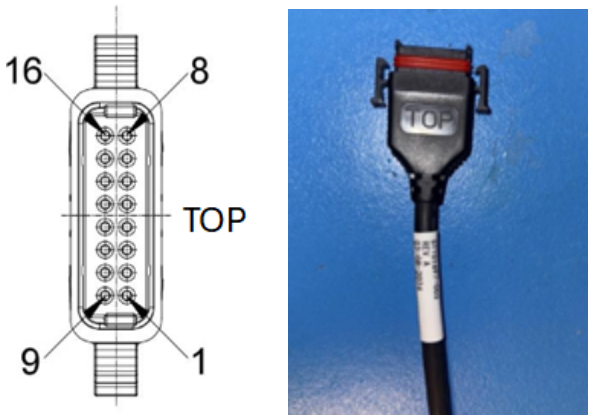


Table 1: Electrical Pin Assignment

PIN	Function	Wire Color
1	I/O_2 OUT	Yellow
2	Ground	Short circuit with pin 10
3	24 V OUT 1	Black
4	Current ADC Input 1 (I IN 0)	Brown
5	RS-485_B	Blue
6	RS-232 TX	Gray
7	Voltage ADC Input 1 (HV Analog 1)	White with a black trace
8	1-Wire I/O	Red with a black trace
9	I/O_1 OUT	Green
10	Ground	Short circuit with pin 2
11	24 V OUT 2	Red
12	Current ADC Input 2 (I IN 1)	Orange
13	RS-485 A	Purple
14	RS-232 RX	White
15	Voltage ADC Input 2 (HV Analog 2)	Brown with a black trace
16	Ground	Orange with a black trace

2.14.2 Configurable I/Os (Advanced Variant)

The advanced variant supports two configurable I/Os:

- Digital input with weak (1 MΩ) pull-down
- Digital input with 30 to 50 K pull-down
- Digital input with 30 to 50 K pull-up
- Analog input with 0 to 3.3 V range
- Digital output – push-pull

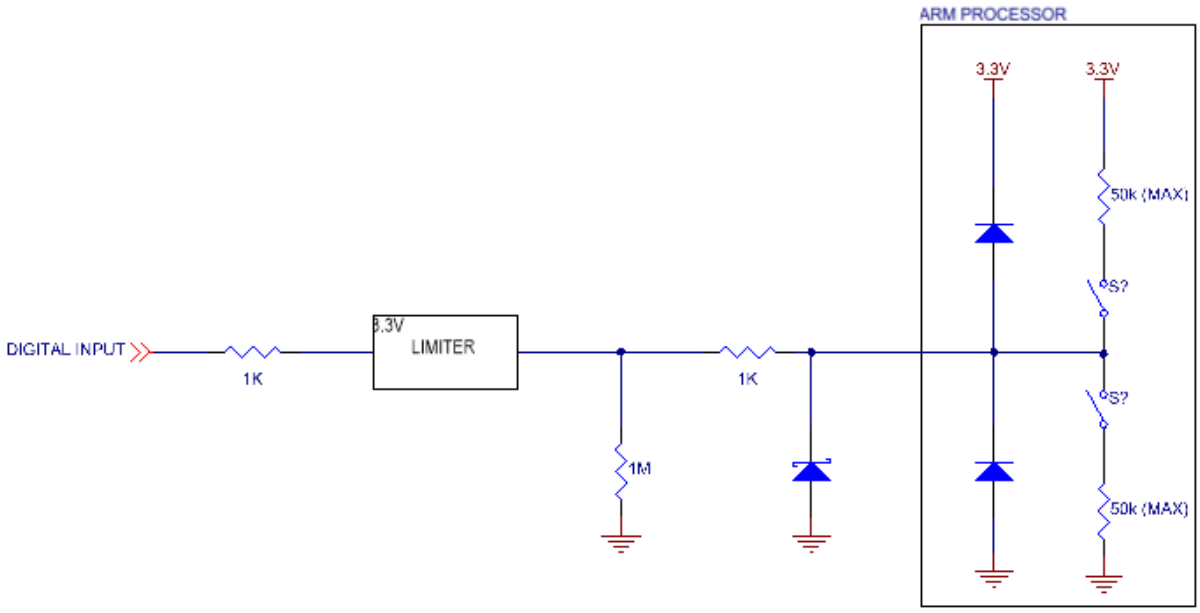
ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

- Digital output – open drain switch-to-ground
- Disabled

2.14.2.1 Digital Input

Figure 3 shows a schematic of the I/O when configured as a digital input.

Figure 3: Digital Input



Input Type	S1	S2
With weak pull-down	Open	Open
With pull-down	Open	Closed
With pull-up	Closed	Open

The input specifications are provided in the table below.

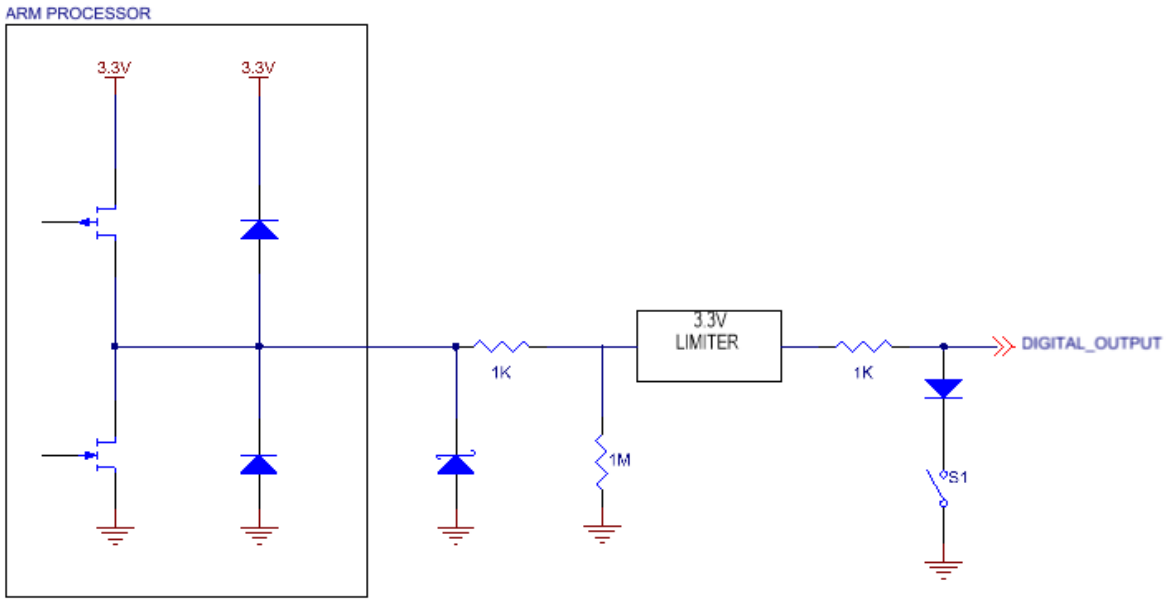
Parameter	Min.	Typical	Max.	Units
Input low range	-10	-	1.15	V
Input high range	2.30	-	200	V
Input current with weak pull-down (weak 1 MΩ pull-down still in place); V_{in} = 3.0 V	-	3	-	µA
Input source current with pull-up (V_{in} = 0.0 V)	-	75	-	µA
Input sink current with pull-down (V_{in} = 3 to 200 V)	-	81	-	µA
Input bandwidth	1	-	-	kHz

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

2.14.2.2 Digital Output

Figure 4 shows a schematic of the I/O when configured as a digital output.

Figure 4: Digital Output



Push-pull	S1 = Open
Open drain	S1 = Closed (Low Impedance) S1 = Open (High Impedance)

2.14.2.2.1 Push-pull

In the push-pull configuration the output is driven directly from the microprocessor.

Parameter	Min.	Typical	Max.	Units
Output high voltage - open circuit	2.8	-	-	V
Output high voltage (sourcing 25 µA)	2.8	-	-	V
Output low voltage (sinking 25 µA)	-	-	0.5	V
Output bandwidth	100	-	-	kHz

2.14.2.2.2 Open Drain

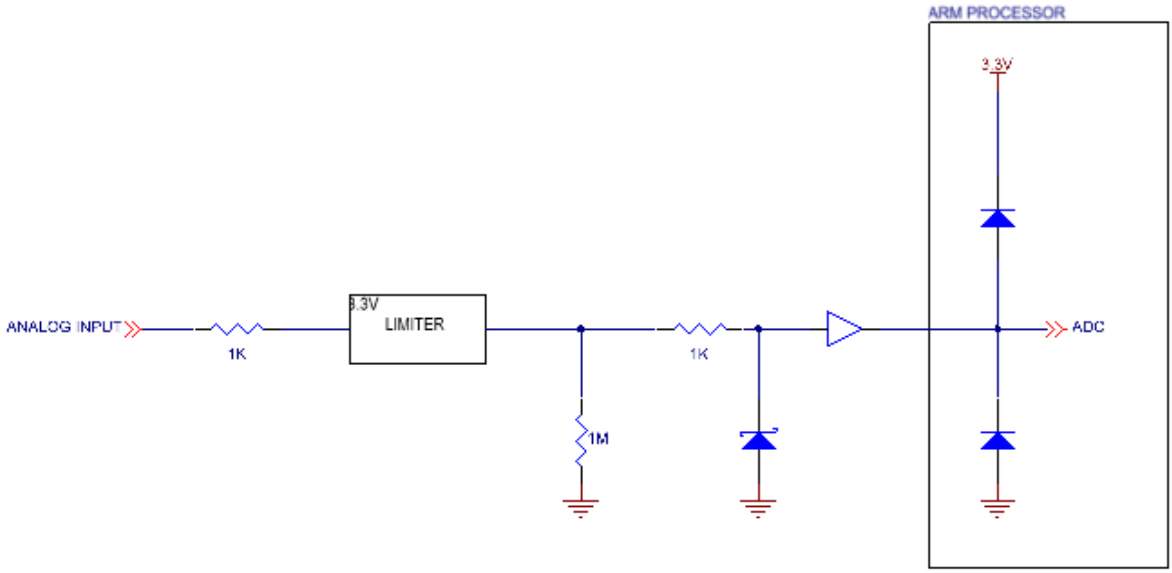
Parameter	Min.	Typical	Max.	Units
Sink current (do not exceed)	-	-	250	mA
Absolute limits (high impedance)	-10	-	150	V
Output voltage	-	1.15	1.35	V
Output bandwidth	100	-	-	kHz

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

2.14.2.3 Analog Input

Figure 5 shows a schematic of the I/O when configured as an analog input.

Figure 5: Analog Input



Parameter	Min.	Typical	Max.	Units
Input impedance	-	1	-	MΩ
Measurement voltage range	0	-	3	V
Resolution (12 bits)	-	0.7	-	mV
Proportional measurement error	-	-	1.5	%
Offset error	-	-	2	LSB
Absolute limits	-10	-	200	V

2.14.3 Serial Interfaces (Advanced Variant)

The ST 8100 has the following interfaces:

- 1 x RS-485
- 1 x RS-232

2.14.3.1 RS-232

The Rx and Tx pins are protected against over- and under-voltage to ±44 V.

The electrical characteristics of the interface are:

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

Parameter	Min.	Typical	Max.	Units
Rx Input Low Threshold for DTE Connected	-	-	-2.7	V
Rx Input High Threshold for DTE Connected	2.7	-	-	V
Rx Threshold for DTE Disconnected	-0.3	-	0.3	V
Serial Rx Input Low Threshold	0.6	-	-	V
Serial Rx Input High Threshold	-	-	2.4	V
Serial Tx Low Output (3 kΩ load)	-	5.4	-	V
Serial Tx High Output (3 kΩ load)	-	5.4	-	V
Over Voltage Protection	-50	-	50	V

2.14.3.2 RS-485

The Rx and Tx pins are protected against over- and under-voltage to ±60 V.

The electrical characteristics of the interface are:

Parameter	Min.	Typical	Max.	Units
Input Common Mode Voltage	-20	-	20	V
Differential Input Threshold	-250	-	-10	mV
Output Common Mode Voltage	-	1.8	3	V
Differential Drive Output, 54 Ω load	1.7	2.3	-	V
ESD Protection (external to driver)				
Human Body Model / Air Gap	-	±8	-	kV
Contact Discharge (IEC 61000-4-2)	-	±15	-	kV

2.14.4 1-Wire (Advanced Variant)

The 1-Wire interface allows connection to downstream 1-Wire devices connected on a bus, or to a single button reader. Relative to any attached 1-Wire device, the transceiver behaves as the master. The 1-Wire driver supports 3 or 3 V devices on the bus.

At standard speed, the 1-Wire supports up to 39 devices over a 61-meter (200 feet) CAT5 cable. In overdrive, the usable expected distance is reduced to ≤15 meters (≤50 feet) with a maximum node count of 9.

The electrical characteristics of the interface are:

Parameter	Min.	Typical	Max.	Units
1-Wire Input High Voltage	1.8	-	-	V
1-Wire Input Low Voltage	-	-	0.6	V
1-Wire Output Low Voltage	-	-	0.2	V
1-Wire Load Current @3V	-	4	-	mA

2.14.5 Open Cover Detection (Advanced Variant)

The advanced variant includes a magnetic switch dedicated to detecting if the cover of the device has been removed.

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

2.14.6 24 V Output (Advanced Variant)

The advanced variant includes two optional 24 V output pins with 30 mA each of output current. The intended use for the 24 V output is to provide power to two external 4 to 20 mA sensors.

2.14.7 High Voltage Analog Input (Advanced Variant)

The advanced variant includes two optional analog voltage inputs. Each input has an accurate input voltage range between 0 and 15 V.

Each analog input of the terminal has an accuracy of 2% over operating temperature range.

2.14.8 Analog Current Input (Advanced Variant)

The advanced variant includes two optional analog current inputs. Each input can support 4 to 20 mA and has an effective resolution of 20-bits.

The analog current inputs are protected from over-voltage, over-current, and are CID2 compliant.

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

3 COMPLIANCE

FCC (United States of America) CFR47 Part 25 and Part 15

- FCC ID: XGS-ST8100
- CONTAINS FCC ID: XPYUBX20VA01
- FCC compliance statement

  This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

ISED (Canada) RSS-170 and ICES-003

- IC: 11881A-ST8100
- CONTAINS IC: 8595A-UBX20VA01
- IC compliance statement:

  This device complies with Industry Canada license-exempt RSS standard(s). Operation is subject to the following two conditions: (1) This device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

  Le present appareil est conforme aux CNR d'Industrie Canada applicable aux appareils radio exempts de licence. L'exploitation est autorisee aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioelectrique subi, meme si le brouillage est susceptible d'en compromettre le fonctionnement.

RF Exposure & Non-modification Warning

RF Exposure statement This equipment complies with FCC and ISED RSS102 radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance of 25 cm (9.8 inches) between the radiator and any part of the human body.

Cet équipement est conforme aux limites d'exposition aux radiations établies par la FCC et ISED RS-102 pour un environnement non contrôlé. Cet équipement doit être installé et utilisé à une distance minimale de 25 cm (9,8 pouces) entre le radiateur et toute partie du corps humain.

Non-modification Warning statement Equipment changes or modifications not expressly approved by the party responsible for FCC compliance, could void the user's authority to operate the equipment and could create a hazardous condition.

CE Mark (Europe) RED 2014/53/EU

- Declaration of Conformity

  Hereby, ORBCOMM Inc. declares that the radio equipment types listed in this document comply with Directive 2014/53/EU.

  The full text of the EU declaration of conformity is available from https://www2.orbcomm.com/eudoc.

Viasat Type Approval

PTCRB (North American certification program for cellular-enabled devices)

IEC 60945 — (pending)

- International Standard that certifies navigation and radio communication equipment and systems for use in maritime application.

FFA (Forum Fisheries Agency of the Pacific Islands) — (pending)

Class I Division 2 (CID2 - (Class I Division 2 as per the US National Electrical Code)) — (pending)

HERO (Hazard of Electromagnetic Radiation to Ordnance) — (pending)

RCM (Australia) — (pending)

ICASA (South Africa) — (pending)

Anatel (Brazil) — (pending)

IFT (Mexico) — (pending)

Environmental

Parameter	Description
Humidity	The device meets all its specifications during exposure to 90% relative humidity at +85°C (185°F), per the test methodology of SAE J1455, section 4.2.3 (8-hour humidity cycle per figure 4a).
Vibration	The device meets all its specifications during exposure to random vehicular vibration levels per SAE J1455, section 4.10.
Mechanical Shock	The device meets all its specifications after exposure to positive and negative saw tooth shock pulses with peaks of 40 G and durations of 11 ms as specified in MIL-STD-810H, section 516.8, Procedure I, section 2.3.2.3.1.
Thermal Shock	The device meets all of its specifications after a thermal shock test as detailed in SAE J1455, section 4.1.3.2.
Handling Drop	The device meets all its specifications after a handling drop test as specified in SAE J1455, section 4.11.3.1
Ingress Protection	Solid Foreign Objects: The enclosure provides protection against ingress of solid foreign objects of IP6X as specified in IEC-60529, section 13.5, Category 2. Water: The enclosure provides protection against ingress of water of IPX7 and IPX9k as specified in IEC-60529.
Salt Spray Atmosphere	The device meets all of its specifications after a salt spray test as detailed in SAE J1455, section 4.3.3.1.
Fungus	The device meets all of its specifications after a fungus test as detailed in SAE J1455, section 4.6.3.
Transit Drop	The device meets all its specifications after transit drop test as specified in SAE J1455, section 4.11.3.2.
ESD	(Enclosure) The device meets all its specifications after exposure of the enclosure to 6 kV ESD contact discharge per IEC 61000-4-2, level 3.

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026

Parameter	Description
Exposure to Chemicals and Oils	The device meets all of its specifications after a light to moderate splash test as detailed in SAE J1455 section 4.4.3.2, for the following chemicals: Window Washer Solvent Gasoline Diesel Fuel Fuel Additives Alcohol Anti-Freeze Water Mixture Degreasers Soap and Detergents Steam Waxes Kerosene Freon Spray Paint Paint Strippers Ether Dust Control Agents (magnesium chloride) Moisture Control Agents (calcium chloride) Ammonia Aluminum brightener (acid wash)

ORBCOMM® PROPRIETARY AND CONFIDENTIAL © Mar 2026